

# Training Day 15 Report – Malware Basics & Static Analysis

**Date:** 02 July 2025

**Module:** Malware Fundamentals & Static Analysis

**Mode:** Theoretical + Practical

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## Main Focus of the Day

Day 16 began the Malware Analysis module by focusing on **malware classification**, **infection techniques**, and **static analysis techniques**. Static analysis involves inspecting a malware sample without executing it.

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## Topics Covered

- **Types of Malware:** Virus, Worm, Trojan, Ransomware, Keylogger, Rootkits, Adware, RATs
  - **Delivery Mechanisms:** Phishing, USB devices, exploit kits, malicious ads (malvertising)
  - **Static Analysis Tools:**
    - `strings` – Identifies readable ASCII/Unicode strings
    - `file` – Determines file type and structure
    - `hashing` – SHA256, MD5 checks for sample integrity
    - `PEStudio` – Portable Executable file inspection
    - `VirusTotal` – Online multi-engine scan report
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## Hands-On Tasks

- Analyzed malware samples in a **safe, isolated VM**
  - Used `strings` command in Linux to extract readable content
  - Uploaded hash values to VirusTotal for detection results
  - Used `PEStudio` to inspect suspicious PE headers and imports
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## Key Takeaways

- Static analysis is **safe and preliminary** but lacks behavioral visibility
- PE headers and string data reveal **indicators of compromise (IOCs)**
- Using VirusTotal helps cross-check with known malware samples

Hash values are **essential for file integrity & threat intelligence**